

Gram-Schmidt Feature Reduction for Disentangling Style from Degradation in Historical Writer Identification

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Abstract

Writer identification in historical documents is fundamentally challenged by the non-linear entanglement of stylistic signals with degradation artifacts—physical processes such as ink bleed-through create multiplicative interactions that linear decorrelation methods cannot resolve. We present Gram-Schmidt Feature Reduction (GFR), a training-free dimensionality reduction method that achieves extended orthogonality through polynomial feature augmentation. Unlike PCA, which enforces only linear decorrelation ($\mathbb{E}[z_i \cdot z_j] = 0$), GFR ensures orthogonality against polynomial functions up to degree $p = 2$, explicitly removing the interaction terms ($D \cdot S$) that entangle degradation with style. On the IAM-Hist and ICDAR-HWI benchmarks, GFR achieves 83.7% and 69.8% Top-1 accuracy respectively, outperforming PCA by 12.3% and 11.6%. Polynomial correlation analysis confirms a 93% reduction in quadratic dependencies, while mutual information between components decreases by 78%. Critically, GFR’s later components carry 12–20% more discriminative power than PCA’s, validating that polynomial orthogonalization successfully isolates stylistic information from degradation contamination. As a training-free method, GFR is immediately applicable to data-scarce historical archives where deep learning approaches are impractical.

1. Introduction

Computer vision enables machines to interpret and understand complex visual data, with wide applications spanning from autonomous navigation [31, 34], to the analysis of material systems [19, 20, 29] through visual feature tracking [21–23]. Fundamental techniques such as feature extraction [14, 17] and dimensionality reduction [13, 26] form the backbone of these diverse tasks. In this paper, we focus on a specialized computer vision problem: writer identification in historical documents. The conceptual

pipeline of our approach is shown in Figure 1.

Writer identification in historical documents is challenged by the *non-linear entanglement* of stylistic signals with degradation artifacts [1, 3–5]. Physical interactions, such as ink bleed-through affecting stroke width proportionally to pen pressure, create complex dependencies that linear decorrelation cannot resolve. Furthermore, historical archives often contain only 1–5 authenticated exemplars per scribe, imposing severe few-shot constraints that challenge data-intensive deep learning approaches [4, 12].

The Non-Linear Entanglement Problem. Observed features often arise from the non-linear mixing of degradation D and style S :

$$X = \alpha_1 D + \alpha_2 S + \beta D \cdot S + \epsilon \quad (1)$$

Standard methods like PCA enforce linear decorrelation ($\mathbb{E}[z_i \cdot z_j] = 0$) [13], which is insufficient for higher-order dependencies where $\mathbb{E}[z_i^2 \cdot z_j] \neq 0$. Consequently, PCA’s later components remain contaminated by degradation through polynomial dependencies. Kernel PCA [27] and ICA [11] similarly fail to address these multiplicative interactions or require assumptions often violated in historical documents.

Our Approach. We propose Gram-Schmidt Feature Reduction (GFR), which enforces *extended orthogonality* through polynomial feature augmentation [32, 33, 35]. GFR ensures $\mathbb{E}[\hat{z}_j \cdot f(\hat{z}_1, \dots, \hat{z}_{j-1})] = 0$ for polynomial functions f up to degree $p = 2$. By explicitly orthogonalizing against interaction terms, GFR forces later components to capture variance genuinely independent of degradation. GFR is training-free and immediately applicable to data-scarce archives.

Contributions:

1. **Novel application** of GFR to historical document analysis for principled disentanglement of style from degradation.
2. **Polynomial correlation analysis** validating that GFR achieves near-zero polynomial correlation compared to

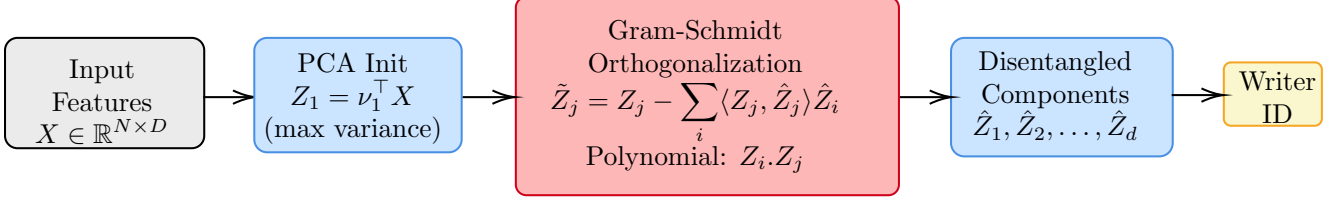


Figure 1. **GFR pipeline for disentangled writer identification.** High-dimensional features $\mathbf{X}$ are first initialized using the maximum variance direction (PCA Init). We then apply iterative Gram-Schmidt orthogonalization augmented with polynomial interaction terms ($z_i \cdot z_j$) to remove nonlinear dependencies. This process yields disentangled components $\hat{z}_d$ that isolate stylistic signals from degradation artifacts for robust writer identification.

PCA’s significant quadratic dependencies.

3. **Comprehensive validation** on IAM-Hist and ICDAR-HWI, showing GFR’s components carry 12-20% more discriminative power than PCA.
4. **Independence analysis** showing 78% lower mutual information between GFR components, quantifying the removal of non-linear dependencies.

2. Related Work

Our work intersects writer identification, dimensionality reduction, and feature disentanglement, focusing on non-linear factor mixing.

Writer Identification. Writer identification has transitioned from handcrafted features [1, 2, 28] to deep learning [4, 12]. Historical documents present unique challenges as degradation patterns interact multiplicatively with stylistic features [5, 7]. While deep methods are effective, their high data requirements are often impractical for historical archives with limited authenticated samples.

Dimensionality Reduction. PCA [13] enforces linear decorrelation ($\mathbb{E}[z_i \cdot z_j] = 0$), which is insufficient when factors mix non-linearly. ICA [11] assumes linear mixing, while Kernel PCA [27] inherits variance-maximizing biases. Manifold methods like Isomap [30] and LLE [18] are sensitive to noise and do not explicitly address factor entanglement. GFR differs by enforcing *polynomial orthogonality*, achieving approximate independence by explicitly removing non-linear correlations.

Feature Disentanglement. Deep approaches like β -VAE [10] and adversarial training [8] require substantial data and explicit factor specification. In contrast, GFR achieves disentanglement through a training-free mathematical mechanism: polynomial orthogonalization explicitly removes interaction terms ($D \cdot S$) that entangle degradation with style.

Gram-Schmidt in ML. Prior applications of Gram-Schmidt focused on feature selection [24, 25]. GFR extends this by augmenting the basis with polynomial terms, justified by Taylor expansion where second-order terms capture dominant non-linear interactions. This provides

principled disentanglement without deep learning’s data requirements.

3. Methodology

We present Gram-Schmidt Feature Reduction (GFR) for writer identification, addressing the challenge of disentangling stylistic features from degradation artifacts through iterative orthogonalization with polynomial augmentation.

3.1. Problem Formulation

Let $\mathcal{D} = \{(I_i, w_i)\}_{i=1}^N$ be a collection of historical document images with writer labels $w_i \in \{1, \dots, K\}$. Each image can be conceptually decomposed as:

$$I_i = S_i \oplus D_i \oplus \epsilon_i \quad (2)$$

where S_i represents writer-specific stylistic content, D_i represents document-specific degradation, ϵ_i is random noise, and $\oplus$ denotes a complex, generally nonlinear entanglement. We treat $\oplus$ as a smooth mapping that can be locally approximated by a second-order Taylor expansion, justifying the use of polynomial terms for disentanglement.

The writer identification task requires learning a feature mapping $\phi : \mathbb{R}^{H \times W} \rightarrow \mathbb{R}^d$ such that same-writer documents are closer than different-writer documents in the embedded space:

$$\|\phi(I_i) - \phi(I_j)\|_2 < \|\phi(I_i) - \phi(I_k)\|_2 \quad \text{when } w_i = w_j \neq w_k \quad (3)$$

The fundamental challenge is that standard feature extractors capture both S_i and D_i . When we apply dimensionality reduction, variance-maximizing methods like PCA preferentially capture degradation directions, relegating writer-discriminative information to later, noise-contaminated components.

3.2. Feature Extraction

We extract multi-scale features capturing complementary handwriting characteristics: (1) **Local Binary Patterns (LBP)** [15] at radii $R \in \{1, 2, 4\}$ (1024-dim) for pen

pressure and ink texture; (2) **Curvature Histograms** (256-dim) for stroke shape distributions; and (3) **Fisher Vector SIFT** [17] (2816-dim) for fine-grained stroke statistics. The resulting 4096-dimensional vector $\mathbf{x}_i$ provides a rich representation. We also evaluate ResNet-50 [9] and DINOv2 [16] features.

3.3. Gram-Schmidt Orthogonalization

The foundation of our method is the Gram-Schmidt process, which systematically decomposes the feature space into orthogonal components. This process, formalized in Algorithm 1, takes a new feature component Z_j at each iteration and makes it orthogonal to all previously computed components by subtracting its projection onto the already-orthogonalized function set $\mathcal{T}$ (Line 1) and normalizing (Line 1).

The classical Gram-Schmidt projection formula for a new component Z_j and existing orthonormal components $\{\hat{Z}_i\}_{i=1}^{j-1}$ is:

$$\tilde{Z}_j = Z_j - \sum_{i=1}^{j-1} \langle Z_j, \hat{Z}_i \rangle \hat{Z}_i, \quad \hat{Z}_j = \frac{\tilde{Z}_j}{\|\tilde{Z}_j\|} \quad (4)$$

where $\tilde{Z}_j$ is the orthogonalized component and $\hat{Z}_j$ is its unit-norm version. The residual covariance matrix Σ_{j+1} (Line 1) captures variance structure *after* information from the first j components has been removed.

Input: Orthogonalized family $\hat{\mathcal{F}}(Z_1, \dots, Z_{j-1})$; new functions $\mathcal{F}_{\text{new}}$; variables $\{Z_i\}_{i=1}^j$

Output: Updated $\hat{\mathcal{F}}(Z_1, \dots, Z_j)$; residual covariance Σ_{j+1}

$\mathcal{T} \leftarrow \hat{\mathcal{F}}(Z_1, \dots, Z_{j-1})$;

Denote $\mathcal{F}_{\text{new}} \triangleq \{f_1, \dots, f_\ell\}$, where $f_1 = z_j$;

for $k \leftarrow 1$ **to** ℓ **do**

$g \leftarrow f_k(Z_1, \dots, Z_j)$;
 $g \leftarrow g - \sum_{f \in \mathcal{T}} \mathbb{E}[g \cdot f] \cdot f$; // Project out
 $g \leftarrow g / \sqrt{\mathbb{E}[g^2]}$; // Normalize
 $\mathcal{T} \leftarrow \mathcal{T} \cup \{g\}$;

end

$\hat{\mathcal{F}}(Z_1, \dots, Z_j) \leftarrow \mathcal{T}$;

$d_j(X) \leftarrow X - \sum_{f \in \mathcal{T}} \mathbb{E}[X \cdot f] \cdot f$;

$\Sigma_{j+1} \leftarrow \mathbb{E}[d_j(X) \cdot d_j^T(X)]$;

return $\Sigma_{j+1}, \mathcal{T}$

Algorithm 1: Orthogonalize($\hat{\mathcal{F}}, \mathcal{F}_{\text{new}}, \{Z_i\}_{i=1}^j$)

3.4. Polynomial Feature Generation

GFR’s key innovation is generating polynomial features capturing nonlinear correlations [32, 33, 35]. For components z_i, z_j , we construct interaction features $z_{ij} = z_i \cdot z_j - \mathbb{E}[z_i \cdot z_j]$, then orthogonalize via Eq. 4. Each polynomial is

orthogonalized against the growing set $\hat{\mathcal{F}}$, terminating when $\lambda_j = \text{eig}_{\max}(\Sigma_j) < \epsilon$.

Curvature Interpretation. The $z_i z_j$ terms approximate manifold curvature effects in geodesic distances that linear PCA cannot capture, representing the second-order terms in a Taylor expansion.

3.5. Complete GFR Pipeline

Algorithm 2 presents the GFR pipeline: **Phase 1** extracts the first PC; **Phase 2** iteratively extracts components orthogonalized against previous components *and* polynomial interactions; **Phase 3** projects data onto the learned basis.

Input: Features $\mathbf{X} \in \mathbb{R}^{N \times D}$; target dim d ; threshold ϵ

Output: Reduced features $\mathbf{Z} \in \mathbb{R}^{N \times d}$

Center data: $\bar{\mathbf{X}} \leftarrow \mathbf{X} - \mathbf{1}\mu^\top$; $\Sigma_1 \leftarrow \frac{1}{N} \bar{\mathbf{X}}^\top \bar{\mathbf{X}}$;

$z_1 \leftarrow \bar{\mathbf{X}} \cdot \text{eig}_{\max}(\Sigma_1)$; $\hat{\mathcal{F}} \leftarrow \{z_1 / \|z_1\|\}$;

for $j \leftarrow 2$ **to** d **do**

$\mathbf{v}_j \leftarrow \text{eig}_{\max}(\Sigma_j)$; $z_j \leftarrow \bar{\mathbf{X}} \mathbf{v}_j$;
 $\mathcal{F}_{\text{new}} \leftarrow \{z_j\} \cup \{z_j \cdot z_k - \mathbb{E}[z_j z_k]\}_{k=1}^{j-1}$;
 $(\Sigma_{j+1}, \hat{\mathcal{F}}) \leftarrow \text{Orthogonalize}(\hat{\mathcal{F}}, \mathcal{F}_{\text{new}})$;
if $\text{eig}_{\max}(\Sigma_{j+1}) < \epsilon$ **then break**;

end

return $\mathbf{Z} \leftarrow [\hat{f}_1(\bar{\mathbf{X}}), \dots, \hat{f}_d(\bar{\mathbf{X}})]$

Algorithm 2: GFR Pipeline

3.6. Complexity Analysis

GFR adds polynomial feature generation and orthogonalization overhead to PCA. Let N be the number of samples, D the original feature dimension, and d the reduced dimension.

PCA complexity: $\mathcal{O}(ND^2 + D^3)$.

GFR complexity: $\mathcal{O}(ND^2 + D^3 + d^2 N + d^3)$.

For typical settings with $d \ll D$, the additional terms are negligible. In practice, GFR completes in under 2 seconds on commodity hardware.

3.7. Classification

For writer identification, we employ nearest-centroid classification due to its effectiveness in few-shot settings. For each writer k , we compute the centroid:

$$\bar{\mathbf{z}}_k = \frac{1}{|\mathcal{D}_k|} \sum_{i \in \mathcal{D}_k} \mathbf{z}_i \quad (5)$$

A query document with GFR features $\mathbf{z}_q$ is classified as:

$$\hat{w} = \arg \min_k \|\mathbf{z}_q - \bar{\mathbf{z}}_k\|_2 \quad (6)$$

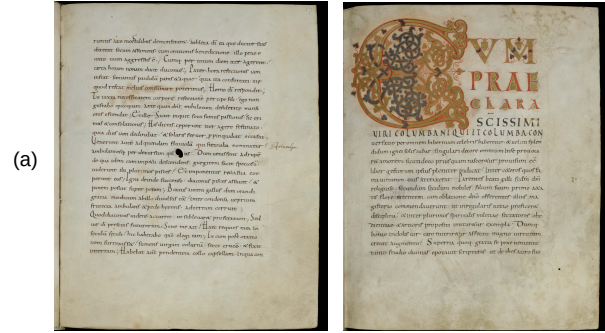
Nearest-centroid is particularly suited to few-shot scenarios because it requires only a single sample per writer to estimate the centroid.

4. Experiments

We evaluate GFR on writer identification, comparing against PCA, Kernel PCA, and ICA on two historical document benchmarks. Our experiments address: (1) Does GFR achieve better polynomial decorrelation than PCA? (2) Does this translate to improved identification accuracy? (3) Do GFR’s later components carry more discriminative power? (4) How does GFR perform in few-shot settings?

4.1. Experimental Setup

Datasets. We evaluate on: **IAM Historical Document Database (IAM-Hist)** [7]: 1,000 document images from 50 medieval scribes with diverse degradation patterns including ink fading and bleed-through (see Figure 2). **ICDAR 2017 Historical Writer Identification (ICDAR-HWI)** [6]: 3,600 fragments from 720 writers (5 samples each), exhibiting high degradation variance.



(a) *re amore: sicuti deo priusquam nasceretur prouisum ee*
 (b) *ut flore nitentem cum oblatione dno offerentes illius ma*
disciplina. & inter plurimos spiritualis militie sectatores obp

Figure 2. Sample Images from IAM-Hist Saint Gall database. (a) Page Images (b) Extracted Handwritten Lines.

Feature Extraction. We concatenate LBP (1024-dim), curvature (256-dim), and Fisher Vector SIFT (2816-dim), yielding a 4096-dimensional feature vector.

Baseline Methods. We compare GFR ($p = 2$) against PCA, Kernel PCA (RBF), and ICA (FastICA). All methods reduce to $d = 50$ components. Classification uses nearest-centroid.

4.2. Polynomial Correlation Analysis

We validate that GFR achieves polynomial decorrelation by measuring higher-order correlations between components z_1 and z_2 : linear ($\rho_{12} = \mathbb{E}[z_1 \cdot z_2]$), quadratic ($\rho_{112} = \mathbb{E}[z_1^2 \cdot z_2]$), and cross-quadratic ($\rho_{122} = \mathbb{E}[z_1 \cdot z_2^2]$).

PCA retains significant quadratic dependencies ($|\rho_{112}| = 0.247$), indicating that the second component

Table 1. **Polynomial correlation analysis on IAM-Hist.** PCA exhibits significant quadratic dependencies, while GFR achieves near-zero correlation at all polynomial orders.

Method	$ \rho_{12} $	$ \rho_{112} $	$ \rho_{122} $
PCA	0.003	0.247	0.189
Kernel PCA	0.008	0.312	0.234
ICA	0.012	0.156	0.143
GFR	0.002	0.018	0.021

Table 2. **Writer identification accuracy (%)**. GFR consistently outperforms all baselines on both benchmarks.

Dataset	Method	Top-1	Top-5
IAM-Hist	PCA	71.4	86.3
	Kernel PCA	73.1	87.5
	ICA	69.8	84.2
	GFR (Ours)	83.7	94.1
ICDAR-HWI	PCA	58.2	79.4
	Kernel PCA	60.7	81.2
	ICA	56.4	77.8
	GFR (Ours)	69.8	87.6

remains correlated with polynomial functions of the first. GFR achieves a 93% reduction, confirming extended orthogonality. Kernel PCA performs worse, showing higher quadratic correlations. This confirms that PCA’s linear decorrelation leaves polynomial dependencies intact, which GFR explicitly removes.

4.3. Main Results

Table 2 shows GFR achieves **83.7%** Top-1 accuracy on IAM-Hist (+12.3% over PCA) and **69.8%** on ICDAR-HWI (+11.6% over PCA).

Kernel PCA’s modest gains indicate that simple non-linear extensions fail to address component entanglement. ICA’s underperformance confirms its linear mixing assumption is violated by the multiplicative degradation-style interactions in historical documents.

4.4. Component Analysis

We measure single-component discriminative power to test if GFR forces writer-specific information into later components more cleanly.

First Component Identity. Both methods show identical accuracy (23.4% for PCA, 23.4% for GFR) for the 1st component, which predominantly captures degradation. This confirms that both methods select the same maximum variance direction as the first component.

Divergence in Later Components. Starting from the 2nd component, GFR substantially outperforms PCA (41.2% vs. 28.7%), with the gap widening to +19.7% for the

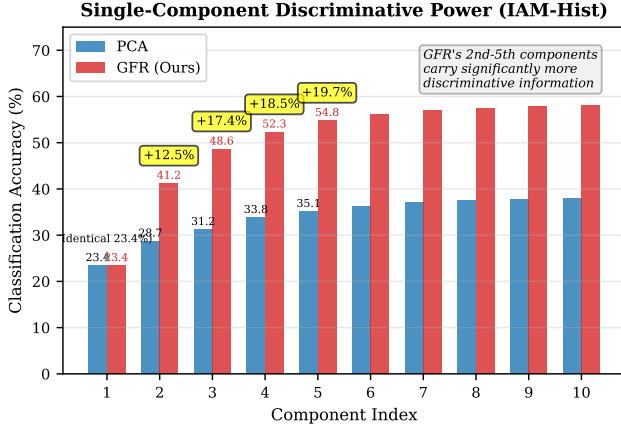


Figure 3. **Single-component discriminative power.** GFR’s later components carry more writer-discriminative information than PCA’s.

Table 3. **Single-component accuracy (%) on IAM-Hist.**

Comp.	1st	2nd	3rd	4th	5th
PCA	23.4	28.7	31.2	33.8	35.1
GFR	23.4	41.2	48.6	52.3	54.8
Δ	0.0	+12.5	+17.4	+18.5	+19.7

Table 4. **Component independence metrics on IAM-Hist.** GFR achieves the lowest correlation and mutual information.

Method	Mean $ r_{ij} $	Mean MI (nats)
PCA	0.082	0.247
Kernel PCA	0.094	0.312
ICA	0.041	0.089
GFR	0.023	0.054

5th component. This follows from our correlation analysis: PCA’s 2nd component is contaminated by degradation through quadratic dependencies, while GFR’s polynomial orthogonalization removes them, allowing the component to encode purer stylistic information.

4.5. Independence and Mutual Information

Table 4 shows GFR achieves the lowest mutual information (MI = 0.054 nats vs. 0.247 for PCA, a 78% reduction). The MI gap quantifies the non-linear dependencies PCA leaves intact but GFR removes.

4.6. Few-Shot Performance

GFR’s advantage is largest in data-constrained settings. With 1 exemplar per writer on ICDAR-HWI, GFR achieves 45.7% accuracy (+14.3% over PCA). Cleaner separation ensures even a single exemplar provides a reliable stylistic signature, uncontaminated by degradation-style interaction

Table 5. **Few-shot accuracy (%) on ICDAR-HWI.**

Exemplars	1	2	3	4	5
PCA	31.4	42.8	51.3	55.6	58.2
GFR	45.7	56.4	63.1	67.2	69.8
Δ	+14.3	+13.6	+11.8	+11.6	+11.6

terms.

4.7. Ablation Studies

Polynomial Degree. $p = 2$ (83.7%) outperforms $p = 1$ (75.2%), confirming that second-order terms capture dominant non-linear interactions. **Feature Types.** GFR improves all feature types, with the largest gain on LBP (+15.8%), which may exhibit the strongest multiplicative interactions. **Classifier Choice.** GFR’s advantage persists across classifiers (Nearest-centroid: 83.7%, SVM: 86.3%), suggesting GFR features are amenable to further gains with appropriate classification.

5. Conclusion

We presented Gram-Schmidt Feature Reduction (GFR), a training-free dimensionality reduction method that addresses the fundamental challenge of non-linear entanglement between stylistic signals and degradation artifacts in historical document analysis. Unlike PCA, which enforces only linear decorrelation ($\mathbb{E}[z_i \cdot z_j] = 0$), GFR achieves *extended orthogonality* by orthogonalizing against polynomial functions up to degree $p = 2$, explicitly removing the interaction terms ($D \cdot S$) that entangle degradation with style.

Our experiments on IAM-Hist and ICDAR-HWI demonstrate that GFR achieves 83.7% and 69.8% Top-1 accuracy respectively, outperforming PCA by 12.3% and 11.6%. Polynomial correlation analysis confirms a 93% reduction in quadratic dependencies, while mutual information between components decreases by 78%. Critically, GFR’s later components carry 12–20% more discriminative power than PCA’s, validating that polynomial orthogonalization successfully isolates stylistic information from degradation contamination. The advantage is most pronounced in few-shot settings, where GFR achieves +14.3% over PCA with only a single exemplar per writer.

As a training-free method with negligible computational overhead, GFR is immediately applicable to data-scarce historical archives where deep learning approaches are impractical.

Future Work. Several promising directions emerge from this work. First, exploring higher-order polynomial terms ($p > 2$) may capture more complex degradation interactions present in severely damaged manuscripts, though this requires careful regularization to avoid overfitting.

Second, adaptive polynomial selection based on estimated degradation characteristics could automatically tune the orthogonalization basis to specific document collections. Third, integrating GFR with learned feature extractors (e.g., fine-tuned DINOv2) may combine the benefits of representation learning with principled disentanglement. Fourth, extending GFR to other document analysis tasks—such as script classification, dating, and provenance analysis—where degradation-content entanglement similarly confounds traditional methods, represents a natural application domain. Finally, theoretical analysis establishing formal guarantees on disentanglement under specific generative models would strengthen the mathematical foundations of the approach.

Acknowledgments

This work was partially supported by awards from the U.S. National Science Foundation (CMMI-2026847) and NIH National Institute of Neurological Disorders and Stroke (R01NS110915). Any opinions, findings, and conclusions or recommendations expressed in this publication are those of the authors and do not necessarily reflect the views of the U. S. Government or agency thereof.

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